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DETERGENT DRAWER ASSEMBLY FOR A WASHING MACHINE APPLIANCE

Abstract

A washing machine appliance includes a cabinet comprising a front panel defining a chamber opening and a top panel, a wash tub positioned within the cabinet and defining a wash chamber, a wash basket rotatably mounted within the wash tub for receiving a load of clothes, and a detergent drawer assembly slidably mounted within the cabinet for receiving a wash additive and directing the wash additive to the wash chamber during operation of the washing machine appliance. The detergent drawer assembly includes a detergent drawer slidably mounted to the cabinet and defining a cartridge chamber, a hinged plate mounted to the detergent drawer and being rotatable between an open position and a closed position, and a cartridge defining a chamber. The cartridge is slidably mounted to the hinged plate when the hinged plate is in the open position.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to washing machine appliances, or more specifically, to the detergent drawer assembly of a washing machine appliance.

BACKGROUND OF THE INVENTION

[0002] Washing machine appliances generally include a tub for containing water or wash fluid, e.g., water and detergent, bleach, and/or other wash additives. A basket is rotatably mounted within the tub and defines a wash chamber for receipt of articles for washing. During normal operation of such washing machine appliances, the wash fluid is directed into the tub and onto articles within the wash chamber of the basket. The basket or an agitation element can rotate at various speeds to agitate articles within the wash chamber, to wring wash fluid from articles within the wash chamber, etc. During a spin or drain cycle, a drain pump assembly may operate to discharge water from within sump.

[0003] Certain conventional washing machine appliances include a detergent drawer assembly for receiving wash additives to be directed to the tub. Notably, detergent drawer assemblies typically include a detergent drawer having chambers within to receive the wash additives and/or water. The chambers occupy accessible space of the detergent drawer, which is the space exposed when the detergent drawer is extended. Notably, there is typically inaccessible space of the detergent drawer, which often corresponds to space covered by the top panel of the washing machine appliance when the detergent drawer is extended. Such inaccessible space, if accessible, could provide additional space within the detergent drawer to be utilized during washing machine appliance operations.

[0004] Accordingly, a washing machine appliance having an improved detergent drawer assembly would be desirable. More specifically, a detergent drawer assembly that is capable of providing access to currently inaccessible space within the detergent drawer would be particularly beneficial.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

[0006] In one exemplary embodiment, a washing machine appliance is provided including a cabinet comprising a front panel defining a chamber opening and a top panel, a wash tub positioned within the cabinet and defining a wash chamber, a wash basket rotatably mounted within the wash tub for receiving a load of clothes, and a detergent drawer assembly slidably mounted within the cabinet for receiving a wash additive and directing the wash additive to the wash chamber during operation of the washing machine appliance. The detergent drawer assembly includes a detergent drawer slidably mounted to the cabinet and defining a cartridge, a hinged plate mounted to the detergent drawer and being rotatable between an open position and a closed position, and a cartridge defining a chamber. The cartridge is slidably mounted to the hinged plate when the hinged plate is in the open position.

[0007] In another exemplary embodiment, a detergent drawer assembly for a washing machine appliance is provided. The washing machine appliance includes a cabinet comprising a top panel, a front panel defining a chamber opening, and a wash tub positioned within the cabinet and defining a wash chamber. The detergent drawer assembly is slidably mounted within the cabinet for receiving a wash additive and directing the wash additive to the wash chamber during operation of the washing machine appliance. The detergent drawer assembly includes a detergent drawer slidably mounted to the cabinet and defining a cartridge chamber, a hinged plate mounted to the detergent drawer and being rotatable between an open position and a closed position, and a cartridge defining a chamber. The cartridge is slidably mounted to the hinged plate when the hinged plate is in the open position.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying

drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a perspective view of a washing machine appliance according to an example embodiment of the present subject matter.

[0011] FIG. 2 provides a front view of the example washing machine appliance of FIG. 1 with a door in the open position according to an example embodiment of the present subject matter.

[0012] FIG. 3 provides a side cross-sectional view of the example washing machine appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0013] FIG. 4 provides another perspective view of the example washing machine appliance of FIG. 1 with a detergent drawer assembly in an extended position according to an example embodiment of the present subject matter.

[0014] FIG. 5 provides another perspective view of the example detergent drawer assembly of FIG. 4 according to an example embodiment of the present subject matter.

[0015] FIG. 6 provides another perspective view of the example detergent drawer assembly of FIG. 4 according to an example embodiment of the present subject matter.

[0016] FIG. 7 provides a perspective view of a hinged plate and a filter cartridge of the example detergent drawer assembly of FIG. 4 with the filter cartridge in the extended position according to an example embodiment of the present subject matter.

[0017] FIG. 8 provides a side cross-sectional view of the example detergent drawer assembly of FIG. 4 according to an example embodiment of the present subject matter.

[0018] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0019] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0020] As used herein, the terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. For example, the approximating language may refer to being within a 10 percent margin.

[0021] Referring now to the figures, FIG. 1 is a perspective view of an exemplary horizontal axis

washing machine appliance **100**, FIG. 2 is a front view of washing machine appliance **100**, FIG. 3 is a side cross-sectional view of washing machine appliance **100**, and FIG. 4 is another perspective view of washing machine appliance **100**. As illustrated, washing machine appliance **100** generally defines a vertical direction V, a lateral direction L, and a transverse direction T, each of which is mutually perpendicular, such that an orthogonal coordinate system is generally defined. Washing machine appliance **100** includes a cabinet **102** that extends between a top **104** and a bottom **106** along the vertical direction V, between a left side **108** and a right side **110** along the lateral direction, and between a front **112** and a rear **114** along the transverse direction T.

[0022] Referring to FIGS. 2 through 4, a wash basket **120** is rotatably mounted within cabinet **102** such that it is rotatable about an axis of rotation A. A motor **122**, e.g., such as a pancake motor, is in mechanical communication with wash basket **120** to selectively rotate wash basket **120** (e.g., during an agitation or a rinse cycle of washing machine appliance **100**). Wash basket **120** is received within a wash tub **124** and defines a wash chamber **126** that is configured for receipt of articles for washing. The wash tub **124** holds wash and rinse fluids for agitation in wash basket **120** within wash tub **124**. As used herein, “wash fluid” may refer to water, detergent, fabric softener, bleach, or any other suitable wash additive or combination thereof. Indeed, for simplicity of discussion, these terms may all be used interchangeably herein without limiting the present subject matter to any particular “wash fluid.”

[0023] Wash basket **120** may define one or more agitator features that extend into wash chamber **126** to assist in agitation and cleaning articles disposed within wash chamber **126** during operation of washing machine appliance **100**. For example, as illustrated in FIG. 3, a plurality of ribs **128** extends from basket **120** into wash chamber **126**. In this manner, for example, ribs **128** may lift articles disposed in wash basket **120** during rotation of wash basket **120**.

[0024] Referring generally to FIGS. 1 through 3, cabinet **102** also includes a front panel **130** which defines a chamber opening **132** that permits user access to wash basket **120** of wash tub **124**. More specifically, washing machine appliance **100** includes a door **134** that is positioned over chamber opening **132** and is rotatably mounted to front panel **130**. In this manner, door **134** permits selective access to chamber opening **132** by being movable between an open position (FIG. 2) facilitating access to a wash tub **124** and a closed position (FIG. 1) prohibiting access to wash tub **124**.

[0025] A window **136** in door **134** permits viewing of wash basket **120** when door **134** is in the closed position, e.g., during operation of washing machine appliance **100**. Door **134** also includes a handle (not labeled) that, e.g., a user may pull when opening and closing door **134**. Further, although door **134** is illustrated as mounted to front panel **130**, it should be appreciated that door **134** may be mounted to another side of cabinet **102** or any other suitable support according to alternative embodiments.

[0026] Referring again to FIG. 3, wash basket **120** also defines a plurality of perforations **140** in order to facilitate fluid communication between an interior of basket **120** and wash tub **124**. A sump **142** is defined by wash tub **124** at a bottom of wash tub **124** along the vertical direction V. Thus, sump **142** is configured for receipt of and generally collects wash fluid during operation of washing machine appliance **100**. For example, during operation of washing machine appliance **100**, wash fluid may be urged by gravity from basket **120** to sump **142** through plurality of perforations **140**.

[0027] A drain pump assembly **144** is located beneath wash tub **124** and is in fluid communication with sump **142** for periodically discharging soiled wash fluid from washing machine appliance **100**. Drain pump assembly **144** may generally include a drain pump **146** which is in fluid communication with sump **142** and with an external drain **148** through a drain hose **150**. During a drain cycle, drain pump **146** urges a flow of wash fluid from sump **142**, through drain hose **150**, and to external drain **148**. More specifically, drain pump **146** includes a motor (not shown) which is energized during a drain cycle such that drain pump **146** draws wash fluid from sump **142** and urges it through drain hose **150** to external drain **148**.

[0028] A spout **152** is configured for directing a flow of fluid into wash tub **124**. For example,

spout **152** may be in fluid communication with a water supply **154** (FIG. 2) in order to direct fluid (e.g., clean water or wash fluid) into wash tub **124**. Spout **152** may also be in fluid communication with the sump **142**. For example, pump assembly **144** may direct wash fluid disposed in sump **142** to spout **152** in order to circulate wash fluid in wash tub **124**.

[0029] As illustrated in FIG. 3, a detergent drawer assembly **200** is slidably mounted within front panel **130**. Detergent drawer assembly **200** receives a wash additive (e.g., detergent, fabric softener, bleach, or any other suitable liquid or powder) and directs the fluid additive to wash tub **124** during operation of washing machine appliance **100**. According to the illustrated embodiment, detergent drawer assembly **200** may also be fluidly coupled to spout **152** to facilitate the complete and accurate dispensing of wash additive.

[0030] In addition, a water supply valve **156** may provide a flow of water from a water supply source (such as a municipal water supply **154**) into detergent drawer assembly **200** and into wash tub **124**. In this manner, water supply valve **156** may generally be operable to supply water into detergent drawer assembly **200** to generate a wash fluid, e.g., for use in a wash cycle, or a flow of fresh water, e.g., for a rinse cycle. It should be appreciated that water supply valve **156** may be positioned at any other suitable location within cabinet **102**. In addition, although water supply valve **156** is described herein as regulating the flow of “wash fluid,” it should be appreciated that this term includes, water, detergent, other additives, or some mixture thereof.

[0031] A control panel **160** including a plurality of input selectors **162** is coupled to front panel **130**. Control panel **160** and input selectors **162** collectively form a user interface input for operator selection of machine cycles and features. For example, in one embodiment, a display **164** indicates selected features, a countdown timer, and/or other items of interest to machine users.

[0032] Operation of washing machine appliance **100** is controlled by a controller or processing device **166** (FIG. 1) that is operatively coupled to control panel **160** for user manipulation to select washing machine cycles and features. In response to user manipulation of control panel **160**, controller **166** operates the various components of washing machine appliance **100** to execute selected machine cycles and features.

[0033] Controller **166** may include a memory and microprocessor, such as a general or special purpose microprocessor operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. Alternatively, controller **166** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software. Control panel **160** and other components of washing machine appliance **100** may be in communication with controller **166** via one or more signal lines or shared communication busses.

[0034] During operation of washing machine appliance **100**, laundry items are loaded into wash basket **120** through chamber opening **132**, and washing operation is initiated through operator manipulation of input selectors **162**. Wash tub **124** is filled with water, detergent, and/or other fluid additives, e.g., via spout **152** and or detergent drawer assembly **200**. One or more valves (e.g., water supply valve **156**) can be controlled by washing machine appliance **100** to provide for filling wash basket **120** to the appropriate level for the amount of articles being washed and/or rinsed. By way of example for a wash mode, once wash basket **120** is properly filled with fluid, the contents of wash basket **120** can be agitated (e.g., with ribs **128**) for washing of laundry items in wash basket **120**.

[0035] After the agitation phase of the wash cycle is completed, wash tub **124** can be drained. Laundry articles can then be rinsed by again adding fluid to wash tub **124**, depending on the particulars of the cleaning cycle selected by a user. Ribs **128** may again provide agitation within

wash basket **120**. One or more spin cycles may also be used. In particular, a spin cycle may be applied after the wash cycle and/or after the rinse cycle in order to wring wash fluid from the articles being washed. During a final spin cycle, basket **120** is rotated at relatively high speeds and drain pump assembly **144** may discharge wash fluid from sump **142**. After articles disposed in wash basket **120** are cleaned, washed, and/or rinsed, the user can remove the articles from wash basket **120**, e.g., by opening door **134** and reaching into wash basket **120** through chamber opening **132**. [0036] While described in the context of a specific embodiment of horizontal axis washing machine appliance **100**, using the teachings disclosed herein it will be understood that horizontal axis washing machine appliance **100** is provided by way of example only. Other washing machine appliances having different configurations, different appearances, and/or different features may also be utilized with the present subject matter as well, e.g., vertical axis washing machine appliances. [0037] Referring to FIG. **1**, a schematic diagram of an external communication system **170** will be described according to an exemplary embodiment of the present subject matter. In general, external communication system **170** is configured for permitting interaction, data transfer, and other communications between washing machine appliance **100** and one or more external devices. For example, this communication may be used to provide and receive operating parameters, user instructions or notifications, performance characteristics, user preferences, or any other suitable information for improved performance of washing machine appliance **100**. In addition, it should be appreciated that external communication system **170** may be used to transfer data or other information to improve performance of one or more external devices or appliances and/or improve user interaction with such devices.

[0038] For example, external communication system **170** permits controller **166** of washing machine appliance **100** to communicate with a separate device external to washing machine appliance **100**, referred to generally herein as an external device **172**. As described in more detail below, these communications may be facilitated using a wired or wireless connection, such as via a network **174**. In general, external device **172** may be any suitable device separate from washing machine appliance **100** that is configured to provide and/or receive communications, information, data, or commands from a user. In this regard, external device **172** may be, for example, a personal phone, a smartphone, a tablet, a laptop or personal computer, a wearable device, a smart home system, or another mobile or remote device.

[0039] In addition, a remote server **176** may be in communication with washing machine appliance **100** and/or external device **172** through network **174**. In this regard, for example, remote server **176** may be a cloud-based server **176**, and is thus located at a distant location, such as in a separate state, country, etc. According to an exemplary embodiment, external device **172** may communicate with a remote server **176** over network **174**, such as the Internet, to transmit/receive data or information, provide user inputs, receive user notifications or instructions, interact with or control washing machine appliance **100**, etc. In addition, external device **172** and remote server **176** may communicate with washing machine appliance **100** to communicate similar information.

[0040] In general, communication between washing machine appliance **100**, external device **172**, remote server **176**, and/or other user devices or appliances may be carried using any type of wired or wireless connection and using any suitable type of communication network, non-limiting examples of which are provided below. For example, external device **172** may be in direct or indirect communication with washing machine appliance **100** through any suitable wired or wireless communication connections or interfaces, such as network **174**. For example, network **174** may include one or more of a local area network (LAN), a wide area network (WAN), a personal area network (PAN), the Internet, a cellular network, any other suitable short- or long-range wireless networks, etc. In addition, communications may be transmitted using any suitable communications devices or protocols, such as via Wi-Fi®, Bluetooth®, Zigbee®, wireless radio, laser, infrared, Ethernet type devices and interfaces, etc. In addition, such communication may use a variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats

(e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

[0041] External communication system **170** is described herein according to an exemplary embodiment of the present subject matter. However, it should be appreciated that the exemplary functions and configurations of external communication system **170** provided herein are used only as examples to facilitate description of aspects of the present subject matter. System configurations may vary, other communication devices may be used to communicate directly or indirectly with one or more associated appliances, other communication protocols and steps may be implemented, etc. These variations and modifications are contemplated as within the scope of the present subject matter.

[0042] Referring now to FIGS. **1** through **8**, the detergent drawer assembly **200** that may be used with the washing machine appliance **100** will be described according to example embodiments of the present subject matter. As explained above, detergent drawer assembly **200** receives the wash additive (e.g., detergent, fabric softener, bleach, or any other suitable liquid or powder). As such, detergent drawer assembly **200** includes a detergent drawer **210** that defines one or more chambers **212** therein for receiving various wash additives and/or water. The detergent drawer **210** is slidably mounted within the front panel **130**. In this manner, the detergent drawer **210** permits selective access to chamber(s) **212** by being movable/slidable between an open or extended position (FIGS. **4-6**) facilitating access to the chamber(s) **212** and a closed or retracted position (FIGS. **1-3, 8**) prohibiting access to the chamber(s) **212**. Notably, as explained above, the chamber(s) **212** occupy the accessible space of the detergent drawer **210**, which is the space exposed when the detergent drawer **210** is extended. However, detergent drawer **210** also has space that is covered by a top panel **138** of the cabinet **102** when the detergent drawer **210** is extended. In conventional detergent drawer assemblies, such space would be inaccessible. Accordingly, however, aspects of the present subject matter are directed to detergent drawer assembly **200** and features for addressing these and other issues experienced with conventional detergent drawer assemblies.

[0043] According to example embodiments, one of the chambers **212** defined by the detergent drawer **210** of the detergent drawer assembly **200** is a cartridge chamber **214** for receiving a cartridge therein, such as a filter cartridge **230** (FIGS. **5-8**) for use in filtering particles from water received within the cartridge chamber **214**. As best illustrated in FIGS. **5** and **6**, the cartridge chamber **214** is positioned at the rear of the detergent drawer **210** in the transverse direction T. As such, when the detergent drawer **210** is in the extended position, a portion of the cartridge chamber **214** is covered by the top panel **138** of the cabinet **102**.

[0044] As best illustrated in FIGS. **5** through **7**, detergent drawer assembly **200** may also include a hinged plate **220**. The hinged plate **220** is rotatably mounted to the detergent drawer **210** for providing selective access to the cartridge chamber **214** and the filter cartridge **230** mentioned above. Notably, contrary to conventional designs, the hinged plate **220** and the filter cartridge **230** of the detergent drawer assembly **200** may facilitate access to the portion of the cartridge chamber **214** that is covered by the top panel **138** of the cabinet **102** which, in conventional detergent drawer assemblies would be inaccessible space.

[0045] The filter cartridge **230** is mounted to the hinged plate **220** and rotates with the hinged plate **220**. The hinged plate **220** is rotatable between an open position (FIGS. **5-6**) facilitating access to the filter cartridge **230** and the cartridge chamber **214** and a closed position (FIG. **4**) in which the filter cartridge **230** is positioned within the cartridge chamber **214**. For example, according to exemplary embodiments, the hinged plate **220** may include a pin **222**. The pin **222** may be received within a pair of drawer bosses **216** defined on or within the detergent drawer **210** and may define a pivot axis P about which the hinged plate **220** is rotatable between the open position and the closed position.

[0046] As best illustrated in FIG. **7**, the filter cartridge **230** defines a filter chamber **232** for receiving a filter (not shown) for filtering particles from water received within the cartridge chamber **214**. The filter cartridge **230** is slidably mounted to the hinged plate **220** when the hinged

plate **220** is the open position. In this manner, the filter cartridge **230** permits selective access to the filter chamber **232** and, thus, the filter (not shown), by being movable/slidable between an open or extended position (FIG. 7) facilitating access to the filter chamber **232** and a closed or retracted position (FIG. 5) in which the hinged plate **220** covers the filter chamber **232** and the filter cartridge **230** is received within the cartridge chamber **214**. For example, according to exemplary embodiments, the filter cartridge **230** may include a pair of flanges **234** for receiving within a pair of flange inserts **224** defined by the hinged plate **220** so the filter cartridge **230** is slidable between the extended position and the retracted position. Furthermore, as best illustrated in FIG. 5, when the filter cartridge **230** is in the retracted position, at least a portion of the filter cartridge **230** is received within the portion of the cartridge chamber **214** covered by the top panel **138** of the cabinet **102**.

[0047] The filter cartridge **230** may be fluidly coupled to a recirculation system and recirculation pump so that the filter (not shown) within the filter chamber **232** receives recirculated wash fluid/water from sump **142** and filters particles from the recirculated wash fluid/water. In this manner, wash fluid/water may be recirculated within washing machine appliance **100** for improved water efficiency and filtered prior to exiting the washing machine appliance **100** for preventing harmful particles from entering the environment.

[0048] As best illustrated in FIG. 8, the wash fluid/water may enter the filter chamber **232** of the filter cartridge **230** via a recirculation nozzle **250** directed into the filter cartridge **230** when the detergent drawer **210** is in the retracted position. In this respect, the recirculation nozzle **250** may be received within a cartridge opening **236** (FIGS. 7-8) defined by the filter cartridge **230** when the detergent drawer **210** is in the retracted position. For example, according to exemplary embodiments, the recirculation nozzle **250** may include an internal conduit **252** received within the cartridge opening **236** for conveying wash fluid/water to the filter chamber **232**. The recirculation nozzle **250** may also include an external conduit **256**. The internal conduit **252** may define a conduit boss **254** for engaging/receiving the external conduit **256** therein when the detergent drawer **210** is in the retracted position and, thus, establishing a fluid coupling/connection between the external conduit **256** and the internal conduit **252** such that wash fluid/water may be conveyed between the internal conduit **252** and the external conduit **256**. Furthermore, the internal conduit **252** may be fixed to the detergent drawer **210** such that the internal conduit **252** moves with the detergent drawer **210** between the extended position and the retracted position. In this respect, as the detergent drawer **210** is moved toward the extended position, the internal conduit **252** disengages or separates from the external conduit **256**.

[0049] As best illustrated in FIGS. 7 and 8, the filter cartridge **230** may include one or more walls **238** each being formed with a screen or mesh filter **240** to assist the filter (not shown) within the filter chamber **232** from filtering particles from the recirculated wash fluid/water. The recirculated wash fluid/water may freely pass through the mesh filter **240** of the wall(s) **238** and into the cartridge chamber **214** of the detergent drawer **210** while the mesh filter **240** separates particles from the recirculated wash fluid/water and contains/traps the particles within the filter chamber **232** of the filter cartridge **230**.

[0050] As best illustrated in FIG. 8, the detergent drawer assembly **200** may also include a drawer chute **260** for directing the filtered recirculated water to the wash tub **124**. The drawer chute **260** may thus be positioned below the cartridge chamber **214** in the vertical direction V for receiving filtered recirculated water from the cartridge chamber **214** via one or more drain holes **218** defined by the bottom of the cartridge chamber **214**. The drawer chute **260** may define a sloped wall **262** such that the filtered recirculated water is directed toward a chute drain **264** and, thus, to the wash tub **124**.

[0051] As explained herein, aspects of the present subject matter are generally directed to a detergent drawer assembly design of a washing machine appliance that includes a rotatable hinged plate and a slidable cartridge chamber. This detergent drawer assembly including the rotatable

hinged plate the slidable cartridge chamber facilitates access to previously inaccessible space of the portion of the detergent drawer contained within the washing machine appliance when the detergent drawer is open/extended. Newly accessible space within the detergent drawer may be utilized for filtering particles from recirculated wash fluid to prevent harmful particles from entering the environment.

[0052] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A washing machine appliance comprising: a cabinet comprising a front panel defining a chamber opening, the cabinet further comprising a top panel; a wash tub positioned within the cabinet and defining a wash chamber; a wash basket rotatably mounted within the wash tub for receiving a load of clothes; and a detergent drawer assembly slidably mounted within the cabinet for receiving a wash additive and directing the wash additive to the wash chamber during operation of the washing machine appliance, the detergent drawer assembly comprising: a detergent drawer slidably mounted to the cabinet and defining a cartridge chamber; a hinged plate mounted to the detergent drawer and being rotatable between an open position and a closed position; and a cartridge defining a chamber, the cartridge being slidably mounted to the hinged plate when the hinged plate is in the open position.
2. The washing machine appliance of claim 1, wherein the cartridge is positioned within the cartridge chamber when the hinged plate is in the closed position.
3. The washing machine appliance of claim 1, wherein, when the hinged plate is in the open position, the cartridge is slidable between an extended position for providing access to the chamber, and a retracted position in which the cartridge is received within the cartridge chamber.
4. The washing machine appliance of claim 1, wherein the detergent drawer is slidable between a retracted position and an extended position in which a portion of the cartridge chamber is covered by the top panel of the cabinet.
5. The washing machine appliance of claim 4, wherein at least a portion of the cartridge is received within the portion of the cartridge chamber covered by the top panel of the cabinet when the cartridge is in the retracted position.
6. The washing machine appliance of claim 4, further comprising: a recirculation nozzle directed into the cartridge when the detergent drawer is in the retracted position.
7. The washing machine appliance of claim 6, wherein the cartridge defines an opening for receiving the recirculation nozzle when the detergent drawer is in the retracted position.
8. The washing machine appliance of claim 1, wherein the cartridge is a filter cartridge and the chamber defined by the cartridge is a filter chamber for receiving a filter for filtering particles from recirculated water.
9. The washing machine appliance of claim 1, wherein the detergent drawer defines a drawer chute for directing filtered water from the chamber to the wash chamber.
10. The washing machine appliance of claim 1, wherein the cartridge rotates with the hinged plate.
11. The washing machine appliance of claim 1, wherein the hinged plate of the detergent drawer assembly comprises: a pin defining a pivot axis about which the hinged plate is rotatable, the pin being received in a pair of bosses defined on the detergent drawer.

- 12.** The washing machine appliance of claim 1, wherein the cartridge of the detergent drawer assembly comprises: at least one wall formed with a mesh filter.
- 13.** A detergent drawer assembly for a washing machine appliance, the washing machine appliance comprising a cabinet comprising a top panel and a front panel defining a chamber opening, the washing machine appliance also comprising a wash tub positioned within the cabinet and defining a wash chamber, the detergent drawer assembly slidably mounted within the cabinet for receiving a wash additive and directing the wash additive to the wash chamber during operation of the washing machine appliance, the detergent drawer assembly comprising: a detergent drawer slidably mounted to the cabinet and defining a cartridge chamber; a hinged plate mounted to the detergent drawer and being rotatable between an open position and a closed position; and a cartridge defining a chamber, the cartridge being slidably mounted to the hinged plate when the hinged plate is in the open position.
- 14.** The detergent drawer assembly of claim 13, wherein the cartridge is positioned within the cartridge chamber when the hinged plate is in the closed position.
- 15.** The detergent drawer assembly of claim 13, wherein, when the hinged plate is in the open position, the cartridge is slidable between an extended position for providing access to the chamber, and a retracted position in which the cartridge is received within the cartridge chamber.
- 16.** The detergent drawer assembly of claim 13, wherein the detergent drawer is slidable between a retracted position and an extended position in which a portion of the cartridge chamber is covered by the top panel of the cabinet.
- 17.** The detergent drawer assembly of claim 16, wherein at least a portion of the cartridge is received within the portion of the cartridge chamber covered by the top panel of the cabinet when the cartridge is in the retracted position.
- 18.** The detergent drawer assembly of claim 16, wherein a recirculation nozzle of the washing machine appliance is directed into the cartridge when the detergent drawer is in the retracted position.
- 19.** The detergent drawer assembly of claim 18, wherein the cartridge defines an opening for receiving the recirculation nozzle of the washing machine appliance when the detergent drawer is in the retracted position.
- 20.** The detergent drawer assembly of claim 13, wherein the hinged plate comprises: a pin defining a pivot axis about which the hinged plate is rotatable, the pin being received in a pair of bosses defined on the detergent drawer.
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